MIL-F-50407A (MU)
15 MAY 1973
SUPERSEDING
MIL-F-50407 (MU)
14 APRIL 1969

MILITARY SPECIFICATION

FUZE, ROCKET, M412A1
ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification contains requirements not normally required by the drawings and provides the quality assurance provisions for the fabrication of parts, assembly and packing of one type of fuze designated as Fuze, Rocket, M412A1 for use in the Rocket, HE, 66MM, Anti-Tank, M72A2.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation, Methods of
MIL-D-1000 - Drawing, Engineering and Associated Lists
MIL-A-2550 - Ammunition and Special Weapons; General Specification for
MIL-D-13684 - Detonator, Electric, M48
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of
MIL-P-14137 - Primer, Stab, M106 Loading, Assembly and Packing

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-331 - Environmental and Performance Tests for Fuze and Fuze Components

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MIL-STD-1167 - Ammunition Data Cards
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

PICATINNY ARSENAL

9243295 - Fuze, Rocket, M412A1 Assembly
8798862 - Carton, Packing, Ammunition for Fuze, Grenade, Rifle, T1014E1 or T1022E2, or Fuze, Rocket, 66MM, M412 without Booster Pellet, Cup Retaining Washer and Cup Disc
8798863 - Box, Packing, Ammunition for Fuze, Grenade, Rifle, T1014E1 or T1022E2, or Fuze, Rocket, 66MM, M412 without Booster Pellet, Cup Retaining Washer and Cup Disc

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. - Materials used in the fabrication of the fuze shall be in accordance with applicable drawings and specifications.

3.1.1 Parts. - All parts used in the assembly of the fuze shall meet all requirements of applicable drawings and specifications.

3.2 Fuze. - The fuze shall comply with all requirements specified on Drawing (dwg.) 9243295 and with all requirements specified in applicable specifications.

3.3 Spring embrittlement. - Springs used in the fuze assembly shall be free from hydrogen embrittlement as evidenced by not breaking when extended to the point of minimum permanent distortion.

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3.4 Porosity of mounting base. - Porosity in the sidewall of the firing pin hole shall not exceed the following requirements:

Area A: Area A is described as the first 0.375 inch of the firing pin hole as measured from the plug end of the base. The total porosity of all voids in this area shall not exceed 0.007 square inch (0.095 diameter).

Area B: Area B is described as a length of 0.095 inch along the length of the firing pin hole adjacent to Area A. The total porosity of all voids in the area shall not exceed 0.0007 square inch (0.030 diameter).

Area C: Area C is described as the remainder of the length of the firing pin hole. The total porosity of all voids in this area shall not exceed 0.0013 square inch (0.040 diameter).

3.5 Electrical Resistance

3.5.1 Setback Assembly

3.5.1.1 Unarmed. - The electrical resistance between the telltale inner contact and the setback support shall not exceed 10 ohms when all leaves are in the unarmed position.

3.5.1.2 Armed. - The electrical resistance between the telltale inner contact and the setback support shall be not less than one megohm when measured with all leaves in the armed position.

3.5.2 Cover Assembly

3.5.2.1 Telltale Rivet. - The electrical resistance between the telltale rivet and the cover shall be not less than one megohm.

3.5.2.2 Uninsulated (ground) terminal. - The electrical resistance between the uninsulated terminal and the cover shall not exceed 0.2 ohm.

3.5.3 Base plate and rotor assembly. - The maximum electrical resistance, indicative of a properly shorted detonator, shall not exceed 0.2 ohm when measured between the rotor contact and the setback support when the assembly is in the unarmed position.

3.5.4 Complete fuze. - The maximum electrical resistance, indicative of an unarmed fuze, shall not exceed ten ohms when measured between the telltale contact in the cover assembly and the ground terminal and shall be not less than one megohm when measured between the ground terminal and insulated terminal in the base connector assembly.

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3.6 Dielectric strength

3.6.1 Cover assembly. - There shall be no breakdown of the insulation between the insulated terminal and the cover when an alternating current potential of 1,000 volts, root mean square, is applied between these points.

3.6.2 Rotor assembly. - There shall be no breakdown of the insulation between the rotor contact assembly and the rotor when an alternating current potential of 1,000 volts, root mean square, is applied between these points.

3.7 Operation of setback assembly

3.7.1 Non-functioning. - The setback assembly, in the base plate and rotor assembly, shall not function when subjected to a minimum arming acceleration of 800 gravities (g's) for a minimum of 5 seconds. Functioning is defined as movement of the leaves sufficient to permit leaf number 1 to clear the rotor pin.

3.7.2 Functioning time. - The functioning time of the setback assembly, in the base plate and rotor assembly, shall not exceed 5.5 milliseconds while the base plate and rotor assembly is being subjected to a maximum arming accleration of 1,100 g's. Functioning time is defined as the interval from the release of leaf number 3 until leaf number 1 has moved to clear the rotor pin.

3.8 Fuze functioning

3.8.1 Graze. - The complete fully armed fuze shall not function the explosive elements when subjected to a minimum accleration of 5 g's for a minimum of 5 seconds, directed towards the booster end of the fuze and along the longitudinal axis of the fuze. The complete fully armed fuze shall function the explosive elements when the accleration is increased to a maximum of 35 g's.

3.8.2 Electrical. - The complete fully armed fuze shall function the explosive elements when the energy from a 0.01 microfarad capacitor, charged to 300 volts maximum, is discharged between the insulated terminal and the cover.

3.9 Rotor arming time

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3.9.1 At plus 70 degress Fahrenheit (F.). - The rotor assembly, in the base plate and rotor assembly, shall rotate from the full safe position to the fully armed position in not less than 70 milliseconds or more than 90 milliseconds.

3.9.2 At minus 40 degrees F. - The rotor assembly, in the base plate and rotor assembly, shall rotate from the full safe position to the fully armed position in not less than 70 milliseconds or more than 100 milliseconds.

3.10 Jolt and jumble. - The fuze shall be subjected to, and must withstand, the jolt test specified in MIL-STD-331, Test 101, with the exception that the fuze shall not be disassembled and inspected after the jolt test. After completion of the jolt test the same fuze shall be subjected to and withstand the jumble test specified in MIL-STD-331, Test 102. None of the place screws shall be completely disassembled as a result of these tests.

3.11 Rough handling. - The fuze shall function satisfactorily after being subjected to the transportation vibration test specified in MIL-STD-331, Test 104 and the five-foot drop test specified in MIL-STD-331, Test 111, except that the fuze shall be dropped bare. Impact shall be on the base end with the base as level as possible. The fuze shall satisfy the criteria for passing test as described in MIL-STD-331.

3.12 Workmanship. - All parts, subassemblies, and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products, and other foreign matter that may impair proper functioning of the fuze or that may cause premature functioning or deterioration in storage. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent used. Painted or coated surfaces shall be restored when these surfaces have been marred or scratched. All required markings shall be neat, clean, and sharply defined.

3.13 First Article Inspection. - This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as

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otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements. Reference shall be made to MIL-STD-109 in order to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product. - At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required by subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Inspection

4.2.1 Submission. - The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of 52 complete fuze assemblies and 30 complete sets of parts, i.e., 30 each of every component and every subsassembly, which have been produced by the contractor or furnished by a supplier and

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which have been manufactured using the same production processes, procedures, and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including statements of findings of tests and examinations and certificates of conformance for materials (see 4.1.1 c). A sample containing known defects will not be submitted, unless specifically authorized by the contracting officer. A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be performed. - Each assembly in the first article sample shall be examined and inspected for every dimension and requirement of the applicable drawings. The entire sample shall be examined in accordance with 4.3.2 and then tested as specified in 4.4.

4.2.3 Rejection. - The first article sample shall be rejected if one or more fuzes, subsassemblies or components fail to conform to any or all of the applicable requirements. The Government reserves the right to terminate its inspection upon any failure of a fuze, subassembly or component in the sample to comply with any of the specified requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article sample or portion thereof. Until a first article sample is accepted, the contractor is, in no way, authorized by the Government to initiate regular production (including procurement of material by the contractor) unless otherwise directed by the Contracting Officer.

4.3 Inspection provisions

4.3.1 Lot formation. - The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and

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inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each lot shall contain: detonators from not more than one interfix lot number from one manufacturer and primers from not more than one interfix lot number from one manufacturer.

4.3.2 Examination. - The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is selected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Spring, Leaf No. 2 (see dwg. 8796414 covering a detail of dwg. 10048781)

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical: None defined			

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Major:	AQL 0.40 percent		
101.	Inside diameter, minimum (min)	Gage	01002
102.	Protective finish incomplete, or inadequate	Visual	01003
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.12)	Visual	01004

4.3.2.2 Spring, Leaf No. 3 (see dwg. 8796415 covering a detail of dwg. 10048781).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Torque at angular displacement	Gage	02001
102.	Inside diameter, min	Gage	02002
103.	Protective finish incomplete or inadequate	Visual	02003
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.12)	Visual	02004

4.3.2.3 Spring, rotor (see dwg. 10048581 covering a detail of dwg. 10048786)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Torque improper at either or both angular displacements	Gage	03001
102.	Inside diameter, min.	Gage	03002
103.	Solid height, maximum (max.)	Gage	03003
104.	Protective finish incomplete or inadequate	Visual	03004
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.12)	Visual	03005

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4.3.2.4 Spring, firing pin (see dwg. 8796380 covering a detail of dwg. 9243295)

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.65 percent		
101.	Solid height, max.	Gage	04001
102.	Load at assembled height	Gage	04002
103.	Protective finish incomplete or inadequate	Visual	04003
Minor:	AQL 1.00 percent		
201.	Ends not ground	Visual	04004
202.	Evidence of poor workmanship (see 3.12)	Visual	04005

4.3.2.5 Pin, firing (see dwg. 8796383 covering a detail of dwg. 9243295)

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.65 percent		
101.	Profile of firing pin point	Gage	05001
102.	Outside diameter, min.	Gage	05002
103.	Inside diameter, max.	Gage	05003
104.	Protective finish incomplete or inadequate	Visual	05004
105.	Radius of locking shaft contact surface	Gage	05007
Minor:	AQL 1.00 percent		
201.	Concentricity of inside diameter with outside diameter	Gage	05005
202.	Evidence of poor workmanship (see 3.12)	Visual	05006

4.3.2.6 Leaf No. 1 (see dwg. 8796408 covering a detail of dwg. 10048781)

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Categories	Defects	Method of Inspection	Code No.
Major:	AQL 0.40 percent		
101.	Plating on inside diameter of hole incomplete or damaged	Visual	06001
102.	Finish improper	Visual	06002
Mjnor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.12)	Visual	06003

4.3.2.7 Leaf No. 2 (see dwg. 8796410 covering a detail of dwg. 10048781)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Plating on inside surface of hole incomplete or damaged	Visual	07001
102.	Finish improper	Visual	07002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.12)	Visual	07003

4.3.2.8 Leaf No. 3 (see dwg. 10048783 covering a detail of dwg. 10048781)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Plating on inside surface of stud hole incomplete or damaged	Visual	08001
102.	Finish improper	Visual	08002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.12)	Visual	08003

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4.3.2.9 Stud, Leaf No. 1 (see dwg. 8796409 covering a detail of dwg. 10048781)

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Plating on intermediate diameter incomplete or damaged	Visual	09001
102.	Finish improper	Visual	09002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.12)	Visual	09003

4.3.2.10 Stud leaf, No. 2 and No. 3 (see dwg. 8796411 and 8796412 covering a detail of dwg. 10048781)

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Plating incomplete or damaged	Visual	10001
102.	Finish improper	Visual	10002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.12)	Visual	10003

4.3.2.11 Base, mounting (see dwg. 10048784 covering a detail of dwg. 10048785)

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 1.00 percent		
101.	Height to cover seating surface, min.	Gage	11001
102.	Height to top of cylindrical portion of base, max.	Gage	11002

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103.	Finish of firing pin cavity improper	Visual	11005
104.	Finish of locking shaft cavity improper	Visual	11006
105.	Diameter of firing pin cavity	Gage	11007
106.	Distance between locking shaft cavity and lever stud cavity	Gage	11008
107.	Diameter of locking shaft cavity	Gage	11009
108.	True position of locking shaft cavity	Gage	11010
109.	Distance from axis of firing pin cavity to axis of locking shaft cavity	Gage	11011
110.	Burrs at intersection of firing pin cavity and locking shaft cavity	Visual	11012
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.12)	Visual	11003

4.3.2.12 Stud Lever (see dwg. 8796400 covering a detail of dwg. 8796399)

Categories	Defect	Method of Inspection	Code No. (see 6.2)
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Diameter of graze lever mounting collar	Gage	12001
102.	Length of graze lever mounting collar	Gage	12002
103.	Concentricity of lever mounting collar with base mounting shaft	Gage	12003
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.12)	Visual	12004

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4.3.2.13 Shaft Locking (see dwg. 8796389 covering a detail of dwg. 10048786)

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical: None defined			
Major:	AQL 0.65 percent		
101.	Outside diameter	Gage	13001
102.	Height of graze lever contact shaft	Gage	13002
103.	Height of firing pin contact shaft	Gage	13003
104.	Length of firing pin contact cutout	Gage	13004
105.	Distance from back of shaft to firing pin cutout	Gage	13005
106.	Distance from back of shaft to graze lever cutout	Gage	13006
Minor:	AQL 0.65 percent		
201.	Finish inside firing pin cutout	Visual	13007
202.	Evidence of poor workmanship (see 3.12)	Visual	13008

4.3.2.14 Lever Graze (see dwg. 8796403 covering a detail of dwg. 10048749)

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical: None defined			
Major:	AQL 0.65 percent		
101.	Thickness, above max.	Gage	14001
102.	Flatness, above max.	Gage	14002
103.	Radius from center of pivot hole to locking shaft contact surface	Gage	14003
104.	Diameter of pivot hole	Gage	14004
105.	Radius of lower corner of locking shaft contact surface	Gage	14005
106.	Height of locking shaft contact surface	Gage	14006
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.12)	Visual	14007

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4.3.2.15 Pallet (see dwg. 8796420 covering a detail of dwg. 10048586)

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical: None defined			
Major:	AQL 0.65 percent		
101.	Distance between pallet teeth	Gage	15001
102.	Distance between outer tooth and axis of pivot hole	Gage	15002
103.	Diameter of pivot hole	Gage	15003
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.12)	Visual	15004

4.3.2.16 Stud, Pallet (see dwg. 8796421 covering a detail of dwg. 10048586).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Rear support mating diameter below min.	Gage	16001
102.	Diameter of pallet mating shaft below min.	Gage	16002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.12)	Visual	16003

4.3.2.17 Gear Pinion (see dwg. 10048602 covering a detail of dwg. 10048586).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical None defined			
Major:	AQL 0.40 percent		
101.	Diameter of rear support mating shaft	Gage	17001
102.	Diameter of star wheel mating shaft .	Gage	17002

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Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.12)	Visual	17003

4.3.2.18 Rear Support Assembly (see dwg. 10048740 covering a detail of dwg. 10048586).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Distance between centers of pallet and starwheel holes	Gage	18001
102.	Diameter of pallet stud hole	Gage	18002
103.	Inside diameter of pinion gear bushing	Gage	18003
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.12)	Visual	18004

4.3.2.19 Pin, Connector (straight) (see dwg. 9251969 covering a detail of dwg. 9251979)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Pin diameter, min. & max.	Gage	19001
102.	Pin length, min.	Gage	19002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.12)	Visual	19003

4.3.2.20 Pin Connector (see dwg. 9251973 covering a detail of dwg. 9251979)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		

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Major:	AQL 0.40 percent		
101.	Pin diameter, min. & max.	Gage	20001
102.	Pin length, min.	Gage	20002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.12)	Visual	20003

4.3.2.21 Base Connector Assembly (see dwg. 9251979 covering a detail of dwg. 9243325)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 1.00 percent		
101.	Contour of connector cavity incorrect	Gage	21001
102.	Connector pin locations improper ..	Gage	21002
103.	Hole locations improper	Gage	21003
104.	Assembly width, max.	Gage	21004
105.	Assembly length, max.	Gage	21005
106.	Assembly height, max.	Gage	21006
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.12)	Visual	21007

4.3.2.22 Cover (see dwg. 9243327 covering a detail of dwg. 9243325)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Location of Base Connector rivet holes improper	Gage	22001
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.12)	Visual	22002

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4.3.2.23 Wheel, Star (see dwg. 8796419 covering a detail of dwg. 10048586).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical: None defined			
Major:	AQL 0.65 percent		
101.	Outside diameter	Gage	23001
102.	Pivot hole diameter	Gage	23002
103.	Tooth profile	Comparator	23003
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.12)	Visual	23004

4.3.2.24 Setback assembly (see dwg. 10048781 covering a detail of dwg. 10048786).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Stud leaf No. 1 or stud leaf No. 2 not peened secured	Visual	24001
2.	Electrical resistance between setback support and telltale inner contact (see 3.5.1)	Test (see 4.4.8)	24007
3.	Distance between the front surface of the setback support and the rear surface of leaf number 1 above max.	Gage	24009
Major:	AQL 0.65 percent		
101.	Ground contact or telltale inner contact loose or deformed	Visual/ Manual	24002
102.	Telltale insulator projects beyond aft surface of telltale inner contact	Visual	24003
103.	Plating damaged (other than peened stud ends)	Visual	24004
104.	Assembly incorrect or incomplete	Visual	24005

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Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.12)	Visual	24006

4.3.2.25 Rotor assembly (see dwg. 10048591 covering a detail of dwg. 10048786)

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Rotor pin fails push test	Test (see dwg.)	25001
Major:	AQL 1.00 percent		
101.	Plating damaged other than stake marks.	Visual	25002
102.	Detonator washer improperly staked	Visual	25003
103.	Rotor contact assembly protrudes beyond outside diameter of rotor	Visual	25004
104.	Rotor gear improperly staked	Visual	25005
105.	Lead wires not taut	Visual	25006
106.	Flash hole between primer cavity and detonator cavity obstructed	Visual	25007
107.	Components missing or loose	Visual/ Manual	25008
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.12)	Visual	25009

4.3.2.26 Base plate and rotor assembly (see dwg. 10048786 covering a detail of dwg. 9243295)

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Detonator not shorted electrically (see 3.5.3)	Test (see 4.4.4)	26001

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Major:	AQL 1.00 percent		
101.	Temporary detonator shorting device not removed	Visual	26003
102.	Primer inverted or loose	Visual/ Manual	26004
103.	Staking inadequate or improperly located	Visual	26005
104.	Components missing or damaged	Visual	26006
105.	Rotor slot not clinched over rotor spring	Visual	26007
106.	Retaining spring improperly positioned, damaged, and / or not under tension (see dwg. 10048785)	Visual/ Manual	26016
107.	Primer staking burrs protrude above outside rotor surface	Visual	26017
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.12)	Visual	26008

4.3.2.27 Cover assembly (see dwg. 9243325 covering a detail of dwg. 9243295)

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Electrical resistance between telltale rivet and cover (see 3.5.2.1)	Test (see 4.4.5)	27007
Major:	AQL 1.00 percent		
101.	Cup not concentric with outside diameter of cover plate	Gage	27001
102.	Outside diameter of cover not concentric with outside diameter of cover plate	Gage	27002
103.	Flash hole obstructed by excessive brazing material	Visual	27003

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104.	Plating incomplete or inadequate ..	Visual	27004
105.	Dielectric strength between terminal and cover (see 3.6.1)	Test (see 4.4.6)	27005
106.	Base connector rivets or telltale rivet loose	Manual	27008
107.	Cover contact damaged or distorted.	Visual	27009
108.	Continuity check between cover and uninsulated terminal (see 3.5.2.2).	Test (see 4.4.5.1)	27010
Minor:	AQL 1.50 Percent		
201.	Depth of booster cup, min.	Gage	27011
202.	Window damaged obscuring vision ...	Visual	27012
203.	Evidence of poor workmanship (see 3.12)	Visual	27013

4.3.2.28 Fuze, Rocket: M412A1 Less Booster Pellet (see dwg. 9243295).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Fuze in armed or partially armed position (see 3.5.4)	Test (see 4.4.7)	28001
2.	One or more electrical test marks missing	Visual (see 4.4.7)	28008
3.	Rotor in armed or partially armed position	Visual (see 4.4.7)	28018
Major:	AQL 0.40 Percent		
101.	Base plug loose or above flush	Visual/ Manual	28002
102.	Assembly dented or damaged	Visual	28003
Minor:	AQL 1.00 Percent		
201.	Marking incorrect, incomplete, illegible or missing	Visual	28004
202.	Evidence of poor workmanship (see 3.12)	Visual	28005

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4.3.2.29 Carton, packing (see dwg. 8798862)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Sealing strip loose, torn, incomplete, wrinkled or improperly adhered	Visual	29001
Minor:	AQL 1.50 percent		
201.	Contents move when shaken	Manual	29003
202.	Label missing	Visual	29004
203.	Marking incorrect, incomplete illegible, or missing	Visual	29005

4.3.2.30 Box, packing (see dwg. 8798863)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Top improperly assembled	Visual/ Manual	30001
102.	Board broken or split	Visual/ Manual	30002
103.	Strapping missing, broken or loose	Visual/ Manual	30003
104.	Contents exposed	Visual	30004
Minor:	AQL 1.00 percent		
201.	Contents loose	Manual	30005
202.	Marking misleading or unidentifiable	Visual	30006

4.3.3 Testing

4.3.3.1 First article sample (see 4.2.1). - The samples submitted as the first article quantity shall be tested as specified herein after

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being examined 100 percent for the applicable classifications of defects in paragraph 4.3.2. Inspection equipment shall be as specified in 4.3.4.

4.3.3.2 Solid film lubricant (see dwg. 8796419, 8796420 and 10048602), Major Defects: Star Wheel - Code No. 31001; Pallet - Code No. 31002; and Pinion Gear - Code No. 31003. - A sample of 30 of each component shall be selected for the first article and tested as specified in 4.4.1. Rejection shall be as specified in 4.2.3.

4.3.3.3 Spring embrittlement (see 3.3), Critical Defects: Spring Leaf No. 2 - Code No. 01005; and Spring Leaf No. 3 - Code No. 02005; Major Defects: Spring Rotor - Code No. 03006 and Spring Firing Pin - Code No. 04006. - A sample of 30 springs shall be selected of each spring type for the first article and tested as specified in 4.4.2. Rejection shall be as specified in 4.2.3.

4.3.3.4 Porosity and surface finish of mounting base (see 3.4 and dwg. 10048784), Porosity: Major Defect - Code No. 11004; and Surface Finish: Major Defect - Code No. 11007. - A sample of 30 mounting bases shall be selected for the first article and tested as specified in 4.4.3. Rejection shall be as specified in 4.2.3.

4.3.3.5 Dielectric strength of rotor assembly (see 3.6.2), Major Defect, Code No. 25010. - A sample of 30 rotor assemblies shall be selected for the first article and tested as specified in 4.4.6.2. Rejection shall be as specified in 4.2.3.

4.3.3.6 Operation of setback assembly (see 3.7); Functioning during non-functioning test, Critical Defect, Code No. 26010; Failure to function within maximum time on functioning time test, Major Defect, Code No. 26011. - A sample of 30 base plate and rotor assemblies shall be selected for the first article and tested as specified in 4.4.10. Rejection shall be as specified in 4.2.3.

4.3.3.7 Surface finish of locking shaft hole of mounting base (see dwg. 10048784), Major Defect, Code No. 11008. - A sample of 30 mounting bases shall be selected for the first article and tested as specified in 4.4.3.1. Rejection shall be as specified in 4.2.3.

4.3.3.8 Functioning of graze elements (see 3.8.1 and dwg. 10048786), Major Defects: Failure during non-functioning or functioning phase of test - Code No. 26009; Improper positioning of locking shaft - Code No. 26002. - A sample of 30 base plate and rotor assemblies shall

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be selected for the first article and assembled with a firing pin, firing pin spring, and base plug to each assembly prior to assembly of the primer in the rotor. The assembly shall then be tested as specified in 4.4.9. Visual examination shall be performed under three power magnification to assure that the locking shaft is bearing properly on the graze lever. Assemblies which contain locking shafts which do not bear on the graze lever as illustrated on the referenced drawing shall be classed defective. Rejection shall be as specified in 4.2.3.

4.3.3.9 Rotor arming time at plus 70 degrees Fahrenheit (see 3.9.1), Arming time, minimum, Critical Defect, Code No. 26012, and arming time, maximum, Major Defect, Code No. 26013 (see dwg. 10048786). A sample of 30 base plate and rotor assemblies shall be selected for the first article and tested as specified in 4.4.11. The temperature conditioning limitations shall be 65°F. to 75°F. Rejection shall be as specified in 4.2.3.

4.3.3.10 Environmental and destructive testing. - A sample of 52 fuzes shall be selected for the first article and subjected to the following tests in the sequence listed. Rejection shall be as specified in 4.2.3.

4.3.3.10.1 Jolt and jumble (see 3.10): Failure to withstand test. - Critical defect. - Code No. 28006; plate screws disassembled. - Major defect - Code No. 28007. - Twelve (12) of the 52 fuzes selected in 4.3.3.10 shall be assembled into closures or simulated closures. The assembly shall then be subjected to the jolt test specified in MIL-STD-331, Test 101 and then without disassembly to the jumble test specified in MIL-STD-331, Test 102. The first article shall be rejected if one or more fuzes have functioned, or if, after disassembly, one or more fuzes are armed or unsafe to handle, or if one or more plate screws have become disassembled.

4.3.3.10.2 Rough Handling (see 3.11). - The remaining 40 fuzes shall be subjected to the transportation vibration test and then without disassembly to the five-foot drop test specified in 3.11.

4.3.3.10.2.1 Arming Tests. - These tests shall be performed on thirty (30) of the forty (40) fuzes only. The cover assembly, graze elements and base plug components shall be removed, but the graze elements (firing pin and firing pin spring) identified for later re-inser - tion into the same fuze and the fuzes shall be subjected to the following tests in the sequence specified.

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TEST	TEST PROCEDURE
Non-Functioning of Setback Assembly	4.4.10
Functioning Time of Setback Assembly	4.4.10
Rotor Arming Time at Plus 70 Degrees F.	4.4.11
Rotor Arming Time at Minus 40 Degrees F.	4.4.11

The temperature tolerance adhered to during performance of the rotor arming time test at plus 70 degrees F and minus 40 degrees F shall be plus and minus five (5) degrees F.

The first article sample shall be rejected if one or more defectives are found according to the defects listed in Table I below. The results may be examined and tested for outliers, but only one outlier is allowed and only for the minus 40 degree F temperature. Refer to 6.6 and 6.7 for calculations of sample standard deviation (sigma) and outliers respectively. After completion of testing (see 4.4.11), the cover assemblies, graze elements and base plugs shall be reassembled to the fuzes and the fuzes tested as follows in 4.3.3.10.2.2 and 4.3.3.10.2.3.

TABLE I

ARMING TEST DEFECTIVES

	DEFECT	CATEGORY	CODE NO.
1.	Functioning during Non-Function Test of Setback Assembly (see 3.7.1)	Critical	28009
2.	Functioning Time of Setback Assembly exceeds maximum (see 3.7.2)	Major	28010
3.	Rotor Arming Time @ 70 Degrees F less than minimum (see 3.9.1)	Critical	28011
4.	Rotor Arming Time @ 70 Degrees F more than maximum (see 3.9.1)	Major	28012
5.	Rotor Arming Time @ Minus 40 Degrees F less than minimum (see 3.9.2)	Critical	28016
6.	Rotor Arming Time @ Minus 40 Degrees F more than maximum (see 3.9.2)	Major	28017

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DEFECT	CATEGORY	CODE NO.
7. Rotor fails to arm @ either temperature	Major	28019
8. Average Rotor Arming Time plus 1.90 Sigma exceeds maximum specified time @ either temperature	Major	28020
9. Average Rotor Arming Time minus 1.90 Sigma is less than the minimum time specified @ either temperature	Major	28021
10. Assembly fails Rough Handling Treatment (see 3.11)	Major	28022

4.3.3.10.2.2 Explosive Graze Functioning (see 3.8.1). - Fifteen (15) of the fuzes previously tested in 4.3.3.10.2.1 and five (5) of the fuzes from 4.3.3.10.2 shall be subjected to the test specified in 4.4.12. The first article sample shall be rejected if one or more fuzes are classed defective according to the defects defined in Table II. Fuzes which do not function shall not be included in the calculations for average graze functioning level and outliers are not permitted. Refer to 6.6 for calculation of sample standard deviation (sigma).

TABLE II

EXPLOSIVE GRAZE FUNCTIONING TEST DEFECTIVES

DEFECT	CATEGORY	CODE NO.
1. Fuze fails to arm @ maximum acceleration level (see 3.7.2)	Major	28019
2. Fuze functions during Non-Functioning Graze Acceleration Test (see 3.8.1)	Major	28013
3. Fuze fails to function explosively or explosive elements fail to perforate the Booster Cup	Major	28015
4. Average graze functioning level plus or minus 1.60 sigma fails to satisfy the specified requirements (see 3.8.1)	Major	28014

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4.3.3.10.2.3 Explosive Electrical Functioning (see 3.8.2). - The remaining twenty (20) fuzes from 4.3.3.10.2 and 4.3.3.10.2.1 shall be subjected to the test specified in 4.4.13. The first article sample shall be rejected if one or more fuzes are classed defective according to the defects listed in Table III.

TABLE III

EXPLOSIVE ELECTRICAL FUNCTIONING TEST DEFECTIVES

	DEFECT	CATEGORY	CODE NO.
1.	Fuze fails to arm @ maximum acceleration level (see 3.7.2)	Major	28019
2.	Fuze fails to function explosively or explosive elements fail to perforate the booster cup	Major	28015

4.3.3.11 Production Testing

4.3.3.11.1 Solid Film Lubricant (see dwgs. 8796419, 8796420 and 10048602) - Major Defects - Star Wheel - Code No. 31001; Pallet - Code No. 31002 and Pinion Gear - Code No. 31003. - A sample of 30 of each component shall be selected from each lot and tested as specified in 4.4.1. The component lot shall be rejected if one or more components are classed defective.

4.3.3.11.2 Spring Embrittlement (see 3.3) - Critical Defects - Spring Leaf No. 2 - Code No. 01005, Spring Leaf No. 3 - Code No. 02005; Major Defects - Spring Rotor - Code No. 03006 and Spring Firing Pin - Code No. 04006. - A sample of 30 springs shall be selected from each spring type included in each heat treatment batch and tested as specified in 4.4.2. If one or more springs are classed defective, the heat treatment batch shall be rejected.

4.3.3.11.3 Porosity and Surface Finish of Mounting Base (see 3.4 and dwg. 10048784) - Porosity - Major Defect - Code No. 11004, Surface Finish - Major Defect - Code No. 11007. - A sample of 30 mounting bases shall be selected from each lot and tested as specified in 4.4.3. If two or more mounting bases fail to conform to the requirements specified, the lot shall be rejected.

4.3.3.11.4 Dielectric Strength of Rotor Assembly (see 3.6.2) - Major Defect - Code No. 25010. - This test shall be performed 100 percent on rotor assemblies prior to assembly of electric detonators. The test shall be performed as specified in 4.4.6.2. Defective assemblies shall be rejected and removed from the lot.

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4.3.3.11.5 Operation of Setback Assembly (see 3.7) - Functioning during Non-Functioning Test - Critical Defect - Code No. 26010; Failure to Function within Maximum Time on Functioning Time Test - Major Defect - Code No. 26011. - This test shall be performed 100 percent on base plate and rotor assemblies. The test shall be performed as specified in 4.4.10. Defective assemblies shall be rejected and removed from the lot.

4.3.3.11.6 Surface Finish of Locking Shaft Hole of Mounting Base (see dwg. 10048784) - Major Defect - Code No. 11008. - A sample of 30 mounting bases shall be selected from each lot and tested as specified in 4.4.3.1. The lot shall be rejected if any base is classed defective.

4.3.3.11.7 Functioning of Graze Elements (see 3.8.1) - Improper positioning of locking shaft - Major Defect - Code No. 26002; Improper functioning of graze elements - Major Defect - Code No. 26009.- This test shall be performed 100 percent on base plate and rotor assemblies, less primer (see dwg. 10048786), by assembling a firing pin, firing pin spring and base plug to each assembly. Visual examination shall be performed under three-power magnification to assure that the locking shaft is bearing properly on the graze lever (see dwg. 10048786). Assemblies which contain locking shafts which do not bear on the graze lever as illustrated on the applicable drawing shall be classed defective and removed from the lot for repositioning of the components necessary to achieve proper assembly. Assemblies which have acceptably passed the above visual examination shall then be subjected to the graze functioning test specified in 4.4.9. Defective assemblies shall be rejected and removed from the lot.

4.3.3.11.8 Rotor Arming Time at Plus 70 Degrees F (see 3.9.1) - Arming Time, Minimum - Critical Defect - Code No. 26012 and Arming Time, Maximum - Major Defect - Code No. 26013 (see dwg. 10048786). - This test shall be performed 100 percent on base plate and rotor assemblies. It is recommended that this test be performed prior to clinching the rotor spring in the rotor shaft to allow for replacement of rotor springs on assemblies which fail the test. The test shall then be repeated on the complete base plate and rotor assemblies (after the clinching operation) to assure that completed assemblies satisfy the arming time requirement. Multiple replacements of the rotor spring are allowed in any individual assembly provided they occur prior to the clinching of the rotor shaft. The test shall be performed as specified in 4.4.11. The temperature conditioning limitations shall be 65°F to 75°F. Defective assemblies shall be rejected and removed from the lot. On completion of arming time test, visually confirm rotor restored to unarmed position - Critical Defect - Code No. 14018 - if any assembly fails visual inspection.

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4.3.3.11.9 Environmental and Destructive Testing. - A sample of 52 fuzes shall be selected from each lot and subjected to the following tests in the sequence listed.

4.3.3.11.9.1 Jolt and Jumble (see 3.10) - Critical Defects - Code No. 16006 and Missing Plate Screws (see 3.10) - Major Defect - Code No. 28007. - Twelve (12) of the fuzes selected in 4.3.3.11.9 shall be assembled into closures or simulated closures. The assemblies shall then be subjected to the jolt test specified in MIL-STD-331, Test 101 and without disassembly to the jumble test specified in MIL-STD-331, Test 102. The lot shall be rejected if one or more fuzes have functioned or if, after disassembly, one or more fuzes are armed or unsafe to handle or if one or more plate screws have become disassembled.

4.3.3.11.9.2 Rough Handling (see 3.11). - The remaining 40 fuzes shall be subjected to the transportation vibration test and then without disassembly to the five-foot drop test specified in 3.11.

4.3.3.11.9.2.1 Arming Tests. - These tests shall be performed on thirty (30) of the forty (40) fuzes only. The cover assembly, graze elements and base plug components shall be removed but the graze elements (firing pin and firing pin spring) identified for later re-insertion into the same fuze and the fuzes shall be subjected to the following tests in the sequence specified.

TEST	TEST PROCEDURE
Non-Functioning of Setback Assembly	4.4.10
Functioning Time of Setback Assembly	4.4.10
Rotor Arming Time at Plus 70 Degrees F	4.4.11
Rotor Arming Time at Minus 40 Degrees F	4.4.11

The temperature tolerance adhered to during performance of the rotor arming time test at plus 70 degrees F and minus 40 degrees F shall be plus and minus 5 degrees F.

The lot shall be rejected if one or more defectives are found according to the defects listed in Table I (see 4.3.3.10.2.1). The results may be examined and tested for outliers, but only one outlier is allowed and only for the minus 40 degree F temperature. Refer to 6.6 and 6.7 for calculations of sample standard deviation (sigma) and outliers respectively. After completion of testing (see 4.4.11), the cover assemblies, graze elements and base plugs shall be reassembled to the fuzes and the fuzes tested as follows in 4.3.3.11.9.2.2 and 4.3.3.11.9.2.3.

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4.3.3.11.9.2.2 Explosive Graze Functioning (see 3.8.1). - Fifteen (15) of the fuzes previously tested in 4.3.3.11.9.2.1 and five (5) of the fuzes from 4.3.3.11.9.2 shall be subjected to the test specified in 4.4.12. The lot shall be rejected if two (2) or more fuzes are classed defective according to the defects listed in Table II (see 4.3.3.10.2.2). Fuzes which do not function shall not be included in the calculations for average graze functioning level. Outliers are not permitted. Refer to 6.6 for calculation of sample standard deviation (sigma).

4.3.3.11.9.2.3 Explosive Electrical Functioning (see 3.8.2). - The remaining twenty (20) fuzes from 4.3.3.11.9.2 and 4.3.3.11.9.2.1 shall be subjected to the test specified in 4.4.13. The lot shall be rejected if two or more fuzes are classed defective according to the defects listed in Table III (see 4.3.3.10.2.3). The lot shall also be rejected if a total of two or more fuzes are classed defective from the total of forty (40) fuzes tested in 4.3.3.11.9.2.2 and 4.3.3.11.9.2.3.

4.3.3.12 Check Test for Possible Deterioration (see MIL-D-13684 and MIL-P-14137 and 6.5). - If the total elapsed time between the original acceptance of any initiator lot and the assembly of the lot into the fuze exceeds two years, or if the initiators have been subjected to adverse conditions for any period of time since previous tests, the initiator lot shall be subjected to and must satisfactorily pass the check test for possible deterioration required by the initiator specification immediately before the initiator lot is assembled into the fuze. These tests shall be performed by the contractor on initiators selected by the Government inspector at the facility assembling the initiators into the fuze. The test shall be conducted at Government expense without cost to the contractor who loaded the initiators or the contractor assembling the initiators into the fuze, and shall not constitute a basis for rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the initiator lot was under the jurisdiction of either contractor.

4.3.3.13 Heat Seal Test of Sealed Bag (see dwg. 8798862) - Major Defect - Code No. 29002. - Sampling, acceptance criteria and test procedures for this test shall be as required in Specification MIL-F-116 (when Level B packing is required).

4.3.4 Inspection Equipment (see 6.2 and 6.3). - The inspection equipment required to perform the examinations and tests prescribed in 4.3.2 is identified under the "Method of Inspection" headings. Unless

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otherwise specified, the inspection equipment used to perform the examinations and tests prescribed in 4.3.3 shall be contractor designed (see 6.3.2).

4.4 Test Methods and Procedures

4.4.1 Solid Film Lubricant. - The component shall be cross-sectioned and the exposed surface polished. The film thickness shall be measured. The component shall be classed defective if bare spots exist on the surfaces or if the thickness of film exceeds the specified drawing requirement.

4.4.2 Spring Embrittlement. - The spring shall be extended in a manner representative of actual functioning (i.e., bending, twisting, tension) until the spring no longer returns to its original condition. The spring shall be classed defective if it breaks prior to this point.

4.4.3 Porosity of Mounting Base. - The mounting base shall be sectioned through the center of the firing pin hole, parallel to the locking shaft hole. The firing pin hole shall be examined for porosity and surface finish. The mounting base shall be classed defective if porosity is in excess of the specified requirements or if the surface finish exceeds the specified drawing requirements.

4.4.3.1 Surface Finish of Locking Shaft Hole of Mounting Base. - The mounting base shall be sectioned through the center of the locking shaft hole and each of the sections shall be examined for surface finish. The mounting base shall be classed defective if it fails to comply with the specified surface finish requirement.

4.4.4 Detonator Short Circuit. - The electrical resistance shall be measured between the rotor contact terminal (10048593) and the setback support (10048782) in the unarmed base plate and rotor assembly (10048786) using Contractor designed test equipment (see 6.3.2). The assembly shall be classed defective if the measured resistance exceeds the specified requirement.

WARNING: The current should be limited to 20 microamperes to prevent inadvertent functioning of the detonators. This test may be dangerous and should be performed behind a suitable barricade.

4.4.5 Insulation Resistance. - The electrical resistance shall be measured between the telltale rivet (10048792) and the cover (9243327) in the cover assembly (9243325) using Contractor designed test equipment (see 6.3.2). The cover assembly shall be classed defective if the resistance measurement is less than the specified requirement.

4.4.5.1 Continuity. - The electrical resistance shall be measured between the straight connector pin (9251969) and the cover (9243327) in

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the cover assembly (9243325) using Contractor designed test equipment (see 6.3.2). The cover assembly shall be classed defective if the resistance measurement exceeds the specified requirement.

4.4.6 Dielectric Strength

4.4.6.1 Cover Assembly. - The specified electrical potential shall be applied between the straight connector pin (9251969) and the cover (9243327) in the cover assembly (9243325) using Contractor designed test equipment (see 6.3.2). Insulation breakdown shall be defined as a leakage current in excess of 20 microamperes. The test equipment shall be capable of detecting the defined breakdown current and shall automatically indicate a defective assembly.

4.4.6.2 Rotor Assembly. - The specified electrical potential shall be applied between the rotor contact assembly (10048592) and the rotor (10048751) in the rotor assembly, less electric detonator, (10048591) using Contractor designed test equipment (see 6.3.2). Insulation breakdown shall be defined as a leakage current in excess of 10 microamperes. The test equipment shall be capable of detecting the defined breakdown current and shall automatically indicate a defective assembly.

4.4.7 Fuze Safety. - The unarmed position of the rotor shaft shall be visually confirmed (see dwg. 9243295). Any fuze containing an armed or partially armed rotor shall be classed defective. After determining that the fuze is visually safe, the electrical resistance shall be measured between the following points: between the telltale rivet (10048792) and the connector pin (9251973) and between the straight connector pin (9251969) and the connector pin (9251973). A red dot shall be readily visible on the cover plate to indicate that the fuze has passed the telltale rivet-to-connector pin electrical test. A black dot shall be readily visible on the cover plate to indicate the fuze has passed the straight connector pin-to-connector pin electrical resistance test. The test equipment shall automatically stamp the proper marking on the fuze, only when each electrical resistance measurement is within the required limits. The fuze shall be classed defective and removed from the lot if, after having been subjected to the test, one or more acceptance marks are not present on the fuze.

WARNING: When measuring the resistances, the test equipment shall limit the current to 20 microamperes to prevent inadvertent functioning of the detonator. This test may be dangerous and should be performed behind a suitable barricade.

4.4.8 Electrical Resistance of Setback Assembly. - This test shall be performed twice: once when the assembly leaves are in the fully safe (unarmed) position and once when the leaves are in the fully armed position. For each test the electrical resistance shall be measured between the telltale inner contact (10048789) and the setback support (10048782) in the setback assembly (10048781) using contractor designed test equipment (see 6.3.2). The setback assembly shall be classed defective if

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either of the resistance measurements fails to meet the specified requirement (3.5.1.1 for the unarmed test and 3.5.1.2 for the armed test).

4.4.9 Graze Functioning. - The assembly shall be examined to assure proper positioning of the locking shaft and graze lever. Prior to testing, the assembly shall provide means to insure against damage to the firing pin and the rotor when the firing pin is released during the functioning phase of this test. The armed assembly shall be mounted on a horizontal centrifuge with the longitudinal axis of the assembly lying along a path of radial acceleration, and the base end pointed towards the center of rotation. The center of gravity is defined as a point 0.50 inches from the base end of the assembly. The minimum non-functioning acceleration shall be applied and maintained for the minimum time period specified. Observation shall be made for non-functioning or functioning. If the assembly has not functioned, the acceleration shall be increased until functioning occurs or until the maximum functioning acceleration is attained. The assembly shall be classed defective if it functions (firing pin is released) during the non-functioning phase or if it fails to function during the functioning phase. Assemblies classed defective may be disassembled and the firing pin (dwg. 8796383) or the firing pin spring dwg. 8796380) or the locking shaft (dwg. 8796389) or any combination thereof may be replaced and retested as often as necessary to obtain acceptable assemblies. Note: Each time the graze elements are reassembled into the base plate and rotor assembly the unit shall be reexamined visually for proper locking shaft position prior to performance of the graze functioning test.

4.4.10 Operation of Setback Assembly

4.4.10.1 Non-Functioning. - The assembly shall be mounted on a horizontal centrifuge with the longitudinal axis of the assembly lying along a path of radial acceleration and the base end pointed away from the center of rotation. The center of gravity is defined as a point 0.35 inches from the base end of the assembly. The minimum non-arming accleration shall be applied for the minimum time specified and observation made for functioning or non-functioning. The assembly shall be classed defective if the setback assembly functions during the non-functioning phase.

4.4.10.2 Functioning Time. - The assembly shall be mounted as in 4.4.10.1. The test equipment shall lock Leaf No. 3 in the fully safe position and the required acceleration applied to the assembly and stabilized. The No. 3 leaf shall be released and the time interval measured from release of leaf No. 3 until leaf No. 1 moves to a position that would release the rotor. The assembly shall be classed defective if the measured time interval exceeds the specified requirement or if the rotor fails to be in the fully armed position after the arming acceleration has been removed.

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WARNING: This test may be dangerous and should be performed behind a suitable barricade.

4.4.11 Rotor Arming Time. - The assemblies shall be conditioned within the specified temperatures for a minimum of three (3) hours. The complete test shall be performed at the required temperature. The arming time (with all leaves in the fully armed position) shall be measured from when the rotor is released from the fully safe position (rotor pin against end of slot) until the rotor reaches the fully armed position, as indicated by the rotor contact making electrical contact with the base contact assembly. The time shall be recorded and the assembly returned to the fully unarmed condition.

WARNING: This test may be dangerous and should be performed behind a suitable barricade.

4.4.12 Explosive Graze Non-Functioning and Functioning. - The assembly shall be placed in the test fixture and subjected to the required arming acceleration. The assembly shall then be observed for proper arming (failures shall be removed without further testing) and reoriented to permit performance of graze testing. The armed assembly shall then be subjected to the minimum non-functioning graze acceleration specified for the minimum time period specified. If functioning does not occur, the acceleration shall be increased until functioning occurs and the functioning g-level shall be recorded to determine compliance with the applicable requirements.

WARNING: This test is dangerous and should be performed behind a suitable barricade.

4.4.13 Explosive Electrical Functioning. - The assembly (fuze) shall be placed in the test fixture and subjected to the required arming acceleration. The assembly shall then be observed for proper arming and failures shall be removed without further testing. The armed fuze shall be tested for electrical functioning by discharging the energy from a 0.01 microfarad capacitor, charged to 300 volts, between the straight connector pin (9251969) and the connector pin (9251973). The fuze shall be classed defective if it fails to arm, if the electric detonator fails to function, or if the booster cup has not been perforated by the functioning of the explosive train.

WARNING: This test may be dangerous and should be performed behind a suitable barricade.

5. PREPARATION FOR DELIVERY

5.1 Preservation, Packaging, Packing and Marking

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5.1.1 Level B. - The fuzes, less booster pellets, shall be packaged, packed and marked in accordance with dwg. 8798862 and 8798863 including the barrier bag listed on 8798862.

5.1.2 Level C

5.1.2.1 Fuzes. - The fuzes, less boster pellet, shall be packaged, packed and marked as specified in 5.1.1 except that barrier bags will not be required.

5.1.2.2 Unmatched, Unassembled Parts. - Cup discs and retaining cup washers shall be packaged, packed and marked in accordance with MIL-STD-1169. If the fuzes are shipped with the graze elements not assembled, the base plugs shall be packaged, packed and marked in accordance with MIL-STD-1169.

6. NOTES

6.1 Ordering Data. - Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Ammunition data cards shall be prepared for each lot in accordance with MIL-STD-1167.

6.2 Inspection Code Numbers. - The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. All submitted designs shall contain a reference to the applicable five-digit code number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. These code numbers are used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. They should also be cited as references on drawings of equipment designs submitted by the contractor (see 6.3.2).

6.3 Inspection Equipment Designs. - Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection" headings in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

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6.3.1 SIE Designs. - SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor Designs. - Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public.) Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment or with SIE designs if not otherwise covered thereby (see 6.3.1.c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five-digit defect code number (or other specific identifying information) from Section 4 of this specification shall be referenced on each contractor design, together with the component or assembly drawing number and revision letter to which the specific design applies.

6.3.3 Submission of Designs for Approval. - Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval to Picatinny Arsenal, ATTN: SMUPA-QA-A-F, Dover, New Jersey 07801 in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.4 Submission of First Article Samples. - Instructions as to the location for the evaluation of the first article shall be obtained from the Contracting Officer. Upon receipt of such request, the Contracting Officer shall advise Picatinny Arsenal and instructions will be issued accordingly. All inquiries should be forwarded to: Commander, Picatinny Arsenal, Dover New Jersey, ATTN: SMUPA-QA-I.

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6.5 Testing of Government Owned Initiators. - The Contracting Officer will arrange to have the fuze contractor reimbursed for expenses incurred in the performance of the check test for deterioration of detonators and primers when the contractor is directed to perform the test.

6.6 Calculations of Sample Standard Deviation. - The sample standard deviation shall be calculated using the formula as follows:

$$S = \sqrt{\frac{\Sigma (X - \bar{X})^2}{n - 1}}$$

where:

S = Sample standard deviation
X = Numerical value for individual observations
$\bar{X}$ = Average value for the sample
n = Sample size

6.7 Calculation of Outliers. - One outlier may be eliminated at the upper limit only when checking the rotor arming time at -40°F. The outlier is defined as follows:

a. Compute $\bar{X}$ and Standard Deviation (S) from data without extreme value (see 6.6)

b. Calculate for K using $K = \frac{\text{High Value} - \bar{X}}{S}$. If K exceeds 2.9, then the high value is an outlier.

NOTICE: When Government drawings, specifications or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied said drawings, specifications or other data is not to be regarded by implication or otherwise in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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Project No. 1340-A-392

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 5535

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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Picatinny Arsenal
Dover, New Jersey 07801

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